Giri's Tech Hub, Pune.

Core Java Machine Test

Batch: June-24 to June-25

Date : 20/02/2025
Time : 08 to 12 Pm.

Instructions:

1. Solve any 12 questions.
2. Input should be from user.
3. Indentation and comments mandatory.
4. Each program 1 marks & all comments 1 marks.
5. Without using any inbuilt functions.

Java :- (Solved Any Six)

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Q1. Write a java program to print this pattern.

```
1
2   *   2
3   *   3   *   3
4   *   4   *   4   *   4
3   *   3   *   3
2   *   2
1
```

Q2. Store N integers in a List. Separate even and odd numbers into two different lists without using built-in filtering.

Input:
6
4 7 10 13 2 9
Output:
Even List: 4 10 2
Odd List: 7 13 9

Description:
Use modulus logic and manually add elements into respective lists.

Q3.Write a java program to find common elements between two arrays using Map.

Input :- Array1: {1, 2, 3, 4, 5}
Array2: {3, 4, 5, 6, 7}
Output :- Common Elements: 3 4 5

Description:
Store first array elements in Map. Traverse second array and check manually if element exists.

Q4. Store N integers in an ArrayList. Rotate the list left by K positions without using Collections.rotate().

Input:
5
1 2 3 4 5
K = 2
Output:
Rotated List: 3 4 5 1 2

Description:
Manually shift elements using loops.

Q5. Write a Java program that takes a string and stores each character with its frequency using a Map.

Input
String: "programming"
Output : p → 1 r → 2 o → 1 g → 2 a → 1 m → 2 i → 1 n → 1

Description:
Do not use any inbuilt string methods for counting. Traverse character by character and update the map manually.

Q6. Store multiple words in an ArrayList. Find the longest word without using compareTo().

Input:
[cat, elephant, dog, tiger]
Output:
elephant

Description:
Compare word lengths manually.

Q7. Store words in an ArrayList. Print pairs of words that are anagrams without using sort().
Input:
[listen, silent, cat, act, dog]
Output:
listen - silent
cat - act
Description:
Compare character frequencies of two words manually.

Q8. Write a java program to find the first repeating word in a sentence.

Input
Sentence: "this is a test this is simple"
Output
First Repeating Word: this

Description :
Manually extract words without split(). Store words in HashMap and find the first word whose frequency

Javascript :- (Solved Any Six)

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Q1. Write a JavaScript program to merge two integer arrays into a single array without duplicates.

Input: arr1 = [1, 2, 3]
arr2 = [2, 3, 4, 5]
Output: [1, 2, 3, 4, 5]

Description:
First copy elements of arr1 into a new array. Then add elements from arr2 only if they do not already exist (check manually using loops).

Q2. Write a JavaScript program to count the frequency of each element in an integer array.

Input:
[2, 3, 2, 5, 3, 2]
Output:
2 → 3 times
3 → 2 times
5 → 1 time

Description:
Use nested loops to count occurrences. Mark visited elements manually.

Q3. Write a JavaScript program to find a continuous subarray whose sum equals a given target value. If such a subarray exists, print the elements of that subarray. Otherwise, print "No Subarray Found".

Input:
Array → [1, 4, 20, 3, 10, 5]
Target → 33
Output:
Subarray → [20, 3, 10]

Description:
Traverse the array and calculate the sum of elements continuously. Use two loops or a sliding window approach to keep adding elements until the sum becomes equal to the target. If the sum exceeds the target, adjust the starting position and continue checking until the required subarray is found. Avoid using inbuilt functions.

Q4. Write a JavaScript program to find the contiguous subarray within an array that has the largest product.

Input: [2, 3, -2, 4]
Output : 6
Explanation : Subarray [2, 3] gives the maximum product = 6.

Description:
Use nested loops to calculate the product of all possible contiguous subarrays. Track the maximum product found. Handle negative numbers carefully while multiplying.

Q5. Write a JavaScript program to reverse each word of a given sentence without changing the word order.

Input : "JavaScript is powerful"
Output : "tpircSavaJ si lufrewop"

Description:
Traverse the string character by character. Reverse characters until a space is found. Store reversed word in a new string. Do not use split() or reverse().

Q6. Write a JavaScript program to remove duplicate characters from a string without using inbuilt methods.

Input : "programming"
Output : "progamin"

Description:
Traverse each character and check manually whether it already exists in a new string before adding it.

Q7. Write a JavaScript program to find the longest palindromic substring in a given string.

Input : "babad"
Output : "bab" (or "aba")

Description:
Check every possible substring using nested loops. For each substring, manually verify whether it is a palindrome by comparing characters from both ends. Track the longest valid palindrome found.

Q8. Write a JavaScript program to perform basic string compression using character counts. If the compressed string length is not smaller than the original string, return the original string.

Input : "aaabbccccd"
Output : "a3b2c4d1"

Description :
Traverse the string character by character. Count consecutive repeated characters manually. Append character + count to a new string. Compare final length with original length without using inbuilt length-reducing functions.

-----ALL THE BEST-----